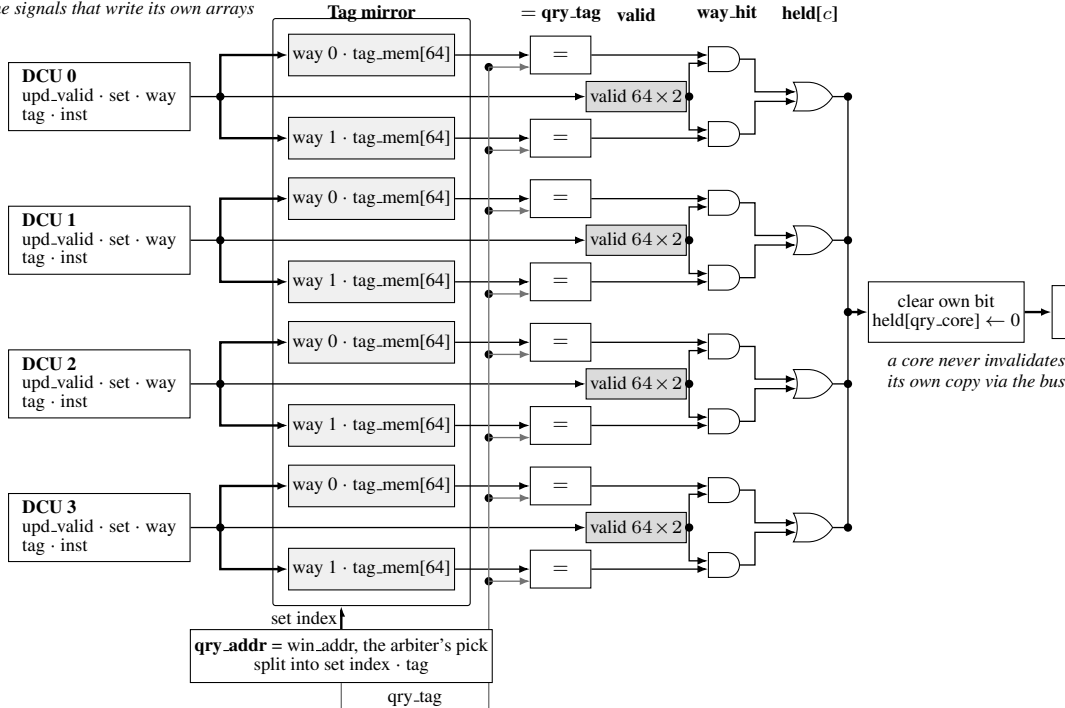


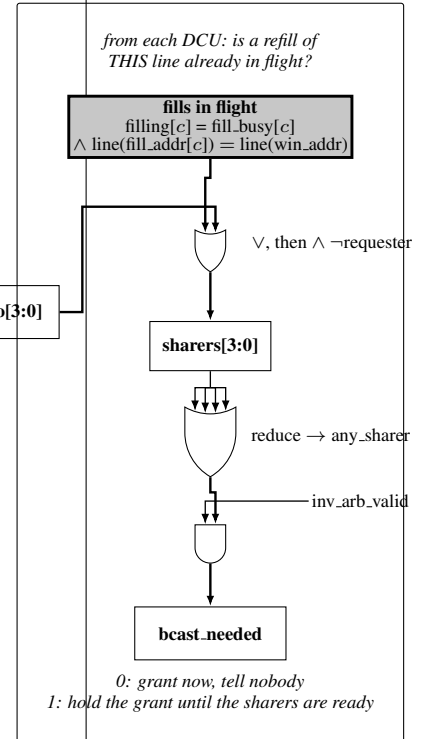
snoop_filter — an exact mirror of every cache's committed tags

one update port per DCU, driven by the same signals that write its own arrays



set index reads all eight tag arrays at once; the tag is checked in all eight comparators
tags: distributed RAM — no reset, one write port, one read port
valid bits: flip-flops — resettable, all 8 read in one cycle

in snoop_bus — residency ∪ imminent residency



distributed RAM (LUTRAM)

flip-flops

the term without which the filter loses invalidations

• junction; a crossing without a dot is not a connection

— qry_tag distribution bus